11/4/24

Parker Lewis

$$y(x+h) = y(x) + hy'(x) + \frac{h^2}{2}y''(x) + \frac{h^3}{6}y'''(x) + O(h^4)$$

$$y'(x) = F(x,y)$$

$$y''(x) = \frac{\partial}{\partial x}\left(F(x,y)\right)$$

$$= \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial x}$$

$$= \frac{\partial F}{\partial x} + y'\frac{\partial F}{\partial y}$$

$$= \frac{\partial F}{\partial x} + F\frac{\partial F}{\partial y} = F_x + FF_y$$

$$y'''(x) = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial x}_{(x,y(x))} + {}_{F(x,y)}\frac{\partial F}{\partial y}\right)$$

$$= \frac{\partial^2 F}{\partial x^2}\frac{\partial x}{\partial x} + \frac{\partial^2 F}{\partial x \partial y}\frac{\partial y}{\partial x} + \frac{\partial}{\partial x}\left(F\frac{\partial F}{\partial y}\right)$$

$$= \frac{\partial^2 F}{\partial x^2}\frac{\partial x}{\partial x} + \frac{\partial^2 F}{\partial x \partial y}\frac{\partial y}{\partial x} + \frac{\partial}{\partial x}\left(F\frac{\partial F}{\partial y}\right)$$

$$= \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial x \partial y}F + \frac{\partial F}{\partial x}\frac{\partial F}{\partial y} + F\frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y}\right)$$

$$= \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial x \partial y}F + \frac{\partial F}{\partial y}\frac{\partial F}{\partial x} + F\left(\frac{\partial F}{\partial y}\right)^2$$

$$+ F\frac{\partial^2 F}{\partial y \partial x}\frac{\partial x}{\partial x} + F\frac{\partial^2 F}{\partial y^2}\frac{\partial y}{\partial x}$$

$$= F_{xx} + F_{xy}F + F_yF_x + FF_y^2$$

$$+ FF_{xy} + F^2F_{yy}$$

$$h\frac{\partial}{\partial h}(hF)\Big|_{h=0}$$

$$= h\left[\frac{\partial h}{\partial h}F + h\frac{\partial F}{\partial h}\right]\Big|_{h=0}$$

$$= h(F)\Big|_{h=0}$$

$$\frac{h^2}{2}\frac{\partial^2}{\partial h^2}(hF)\Big|_{h=0}$$

$$= \frac{h^2}{2}\frac{\partial}{\partial h}\left(\frac{\partial}{\partial h}(hF)\right)\Big|_{h=0}$$

$$= \frac{h^2}{2}\frac{\partial}{\partial h}\left(F + h\frac{\partial F}{\partial h}\right)\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{\partial F}{\partial h} + \frac{\partial}{\partial h}(hF_h)\right]\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{\partial F}{\partial h} + F_h + h\frac{\partial F_h}{\partial h}\right]\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{\partial F}{\partial h} + \frac{\partial F}{\partial h} + h\frac{\partial^2 F}{\partial h^2}\right]\Big|_{h=0}$$

$$h\frac{\partial}{\partial h}(hF)\Big|_{h=0}$$

$$= h\left[\frac{\partial h}{\partial h}F + h\frac{\partial F}{\partial h}\right]\Big|_{h=0}$$

$$= h(F)\Big|_{h=0}$$

$$\frac{h^2}{2}\frac{\partial^2}{\partial h^2}(hF)\Big|_{h=0}$$

$$= \frac{h^2}{2}\frac{\partial}{\partial h}\left(\frac{\partial}{\partial h}(hF)\right)\Big|_{h=0}$$

$$= \frac{h^2}{2}\frac{\partial}{\partial h}\left(F + h\frac{\partial F}{\partial h}\right)\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{\partial F}{\partial h} + \frac{\partial}{\partial h}(hF_h)\right]\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{\partial F}{\partial h} + F_h + h\frac{\partial F_h}{\partial h}\right]\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{\partial F}{\partial h} + \frac{\partial F}{\partial h} + h\frac{\partial^2 F}{\partial h^2}\right]\Big|_{h=0}$$

$$\frac{h^2}{2}\left[\frac{\partial F}{\partial h} + \frac{\partial F}{\partial h} + h\frac{\partial^2 F}{\partial h^2}\right]\Big|_{h=0}$$

$$= \frac{h^2}{2}\left[\frac{2\partial F}{\partial h}\right]\Big|_{h=0}$$

$$\frac{h^3}{6}\frac{\partial^3}{\partial h^3}\left(hF\right)\Big|_{h=0}$$

$$= \frac{h^3}{6}\frac{\partial^2}{\partial h^2}\left(\frac{\partial}{\partial h}\left(hF\right)\right)\Big|_{h=0}$$

$$= \frac{h^3}{6}\frac{\partial}{\partial h}\left(\frac{\partial}{\partial h}\left(\underbrace{\frac{\partial}{\partial h}\left(hF\right)}_{F + h\frac{\partial F}{\partial h}}\right)\right)\Big|_{h=0}$$

$$\underbrace{\qquad\qquad\qquad\qquad}_{2\frac{\partial F}{\partial h} + h\frac{\partial^2 F}{\partial h^2}}$$

$$= \frac{h^3}{6}\frac{\partial}{\partial h}\left(2\frac{\partial F}{\partial h} + h\frac{\partial^2 F}{\partial h^2}\right)\Big|_{h=0}$$

$$y(x+h) = y(x) + hF + \frac{h^2}{2}\left(F_x + FF_y\right)$$
$$+ \frac{h^3}{6}\left(F_{xx} + 2F_{xy}F + F_yF_x + FF_y^2 + F^2F_{yy}\right)$$
$$+ O(h^4)$$

$$y(x+h) = y(x) + \alpha_0 k_0 + \alpha_1 k_1 + \alpha_2 k_2 + O(h^4)$$

$$k_0 = hF(x,y) \checkmark$$
$$k_1 = hF(x+\mu_1 h,\ y+\lambda_1 k_0)$$
$$k_2 = hF(x+\mu_2 h,\ y+\lambda_2 k_1)$$

$$k_1 = hF(x+\mu_1 h,\ y+\lambda_1 k_0) \approx$$

$$hF(x+\mu_1 h,\ y+\lambda_1 k_0)\big|_{h=0} + h\frac{\partial}{\partial h}\left(hF(x+\mu_1 h,\ y+\lambda_1 k_0)\right)\Big|_{h=0}$$

$$+ \frac{h^2}{2}\frac{\partial^2}{\partial h^2}\left(hF(x+\mu_1 h,\ y+\lambda_1 k_0)\right)\Big|_{h=0}$$

$$+ \frac{h^3}{6}\frac{\partial^3}{\partial h^3}\left(hF(x+\mu_1 h,\ y+\lambda_1 k_0)\right)\Big|_{h=0}$$
$$+ O(h^4)$$

$$\frac{h^3}{6}\left[2\frac{\partial^2 F}{\partial h^2} + \frac{\partial^2 F}{\partial h^2} + h\frac{\partial^3 F}{\partial h^3}\right]\Big|_{h=0}$$

$$= \frac{h^3}{6}\left[3\frac{\partial^2 F}{\partial h^2}\right]$$

$$k_1 \approx hF(x,y) + \frac{h^2}{2}\left[2\frac{\partial F}{\partial h}(x+\alpha_1 h, y+\lambda_1 hF)\right]_{h=0}$$

$$+ \frac{h^3}{6}\left[3\frac{\partial^2 F}{\partial h^2}(x+\alpha_1 h, y+\lambda_1 hF)\right]_{h=0} + O(h^4)$$

$$\frac{\partial F}{\partial h}\Big|_{h=0} = \left(\frac{\partial F}{\partial(x+\alpha_1 h)}\frac{\partial(x+\alpha_1 h)}{\partial h} + \frac{\partial F}{\partial(y+\lambda_1 hF)}\frac{\partial(y+\lambda_1 hF)}{\partial h}\right)_{h=0}$$

$$= \left(\alpha_1\frac{\partial F}{\partial(x+\alpha_1 h)} + \lambda_1 F\frac{\partial F}{\partial(y+\lambda_1 Fh)}\right)_{h=0}$$

$$= \alpha_1 F_x + \lambda_1 F_y F$$

$$\frac{\partial^2 F}{\partial h^2}\Big|_{h=0} = \left[\frac{\partial}{\partial h}\left(\frac{\partial F}{\partial h}\right)\right]_{h=0}$$

$$= \frac{\partial}{\partial h}\left(M_1 \frac{\partial F}{\partial (x+M_1 h)} + \lambda_1 F \frac{\partial F}{\partial (y+\lambda_1 F h)}\right)_{h=0}$$

$$= \left(M_1 \frac{\partial^2 F}{\partial^2 (x+M_1 h)} \frac{\partial (x+M_1 h)}{\partial h} + M_1 \frac{\partial^2 F}{\partial (x+M_1 h)\partial (y+\lambda_1 h F)}\right.$$

$$\times \frac{\partial (y+\lambda_1 h F)}{h} + \lambda_1 F \frac{\partial^2 F}{\partial^2 (y+\lambda_1 F h)} \frac{\partial (y+\lambda_1 F h)}{\partial h}$$

$$\left. + \lambda_1 F \frac{\partial^2 F}{\partial (y+\lambda_1 F h)\partial (x+M_1 h)} \frac{\partial (x+M_1 h)}{\partial h}\right)_{h=0}$$

$$= M_1^2 F_{xx} + M_1 \lambda_1 F F_{xy} + \lambda_1^2 F^2 F_{yy}$$

$$+ \lambda_1 F M_1 F_{yx}$$

$$= M_1^2 F_{xx} + \lambda_1^2 F^2 F_{yy} + 2\lambda_1 F M F_{xy}$$

$$k_1 \approx hF + \frac{h^2}{2}\left[2\mu_1 F_x + 2\lambda_1 F_y F\right]$$
$$+ \frac{h^3}{6}\left[3\mu_1^2 F_{xx} + 3\lambda_1^2 F^2 F_{yy} + 6\lambda_1 F \mu_1 F_{xy}\right]$$
$$+ O(h^4)$$

$$k_2 \approx \cancel{(hF)|_{h=0}} + h'(F)|_{h=0}$$
$$+ \frac{h^2}{2}\left(2\frac{\partial F}{\partial h}\right)\Big|_{h=0} + \frac{h^3}{6}\left(3\frac{\partial^2 F}{\partial h^2}\right)\Big|_{h=0}$$
$$+ O(h^4)$$

$$k_2 \approx h(F)|_{h=0} + \frac{h^2}{2}\left(2\frac{\partial F}{\partial h}\right)\Big|_{h=0} + \frac{h^3}{6}\left(3\frac{\partial^2 F}{\partial h^2}\right)\Big|$$

$$\frac{\partial F}{\partial h} = \left(\frac{\partial F}{\partial (x+u_2 h)}\frac{\partial (x+u_2 h)}{\partial h} + \frac{\partial F}{\partial (y+\lambda_2 k_1)}\frac{\partial (y+\lambda_2 k_1)}{\partial h}\right)_{h=0}$$

$$= \frac{\partial F}{\partial (x+u_2 h)} u_2 + \frac{\partial F}{\partial (y+\lambda_2 k_1)} \lambda_2 \left(F + h\frac{\partial F}{\partial h} + \right.$$

$$- h\left[\cancel{2\mu_1 F_x + 2\lambda_1 F_y F}\right] + \frac{h^2}{2}\frac{\partial}{\partial h}\left(\cancel{2\mu_1 F_x + 2\lambda_1 F_y F}\right)$$

$$+ \frac{h^2}{2}\left[\cancel{3\mu_1^2 F_{xx} + 3\lambda_1^2 F^2 F_{yy} + 6\lambda_1 F \mu_1 F_{xy}}\right] + \frac{h^3}{6}\frac{\partial}{\partial h}\left[\cancel{3\mu_1^2 F_{xx} + 3\lambda_1^2 F^2 F_{yy} + 6\lambda_1 F \mu_1 F_{xy}}\right]$$

$$\partial(x+u_2 h)(y+\lambda_2 k_1) \qquad \partial h \qquad \partial h$$

$$\frac{\partial F}{\partial h}\Big|_{h=0} = \left(\frac{\partial E}{\partial x} u_2 + \frac{\partial E}{\partial y} \lambda_2 F\right)\Big|_{h=0}$$

$$\frac{\partial^2 F}{\partial h^2} = \frac{\partial}{\partial h}\left(\frac{\partial F}{\partial (x+u_2 h)} \frac{\partial (x+u_2 h)}{\partial h} + \frac{\partial F}{\partial (y+\lambda_2 k_1)} \frac{\partial (y+\lambda_2 k_1)}{\partial h}\right)$$

$$= \frac{\partial^2 F}{\partial (x+u_2 h)^2} \frac{\partial (x+u_2 h)}{\partial h} \frac{\partial (x+u_2 h)}{\partial h}$$

$$+ \frac{\partial^2 F}{\partial (x+u_2 h)\partial (y+\lambda_2 k_1)} \frac{\partial (y+\lambda_2 k_1)}{\partial h} \frac{\partial (x+u_2 h)}{\partial h}$$

$$+ \frac{\partial^2 F}{\partial (y+\lambda_2 k_1)^2} \frac{\partial (y+\lambda_2 k_1)}{\partial h} \frac{\partial (y+\lambda_2 k_1)}{\partial h}$$

$$+ \frac{\partial^2 F}{\partial (x+u_2 h)\partial (y+\lambda_2 k_1)} \frac{\partial (x+u_2 h)}{\partial h} \frac{\partial (y+\lambda_2 k_1)}{\partial h}$$

$$= u_2^2 \frac{\partial^2 F}{\partial (x+u_2 h)^2}\Big|_{h=0} + u_2 \frac{\partial^2 F}{\partial (x+u_2 h)\partial (y+\lambda_2 k_1)} \frac{\partial (y+\lambda_2 k_1)}{\partial h}$$

$$+ \frac{\partial^2 F}{\partial (y+\lambda_2 k_1)^2} \frac{\partial (y+\lambda_2 k_1)}{\partial h} \frac{\partial (y+\lambda_2 k_1)}{\partial h} +$$

$$\frac{\partial^2 F}{\partial (x+u_2 h)(y+\lambda_2 k_1)} \frac{\partial (x+u_2 h)}{\partial h} \frac{\partial (y+\lambda_2 k_1)}{\partial h}$$

$$\left.\frac{\partial k_1}{\partial h}\right|_{h=0} = F(x,y)$$

* since we did term earlier on k_1 terms *

$$\left.\frac{\partial^2 F}{\partial h^2}\right|_{h=0} = u_2^2 F_{xx} + u_2 F_{xy} F\lambda_2 + F_{yy} F^2 \lambda_2^2 + F_{xy} u_2 \lambda_2 F$$

$$= u_2^2 F_{xx} + F^2 \lambda_2^2 F_{yy} + 2\lambda_2 F u_2 F_{xy}$$

$$k_2 \approx hF + \frac{h^2}{2}\left(2F_x u_2 + 2F_y \lambda_2 F\right) + \frac{h^3}{6}\left(3u_2^2 F_{xx} + 3F^2\lambda_2^2 F_{yy} + 6\lambda_2 F u_2 F_{xy}\right) + O(h^4)$$

$$y(x+h) \approx$$

$$y(x) + \alpha_0 hF + \alpha_1 hF + \frac{\alpha_1 h^2}{2}\left[2u_1 F_x + 2\lambda_1 F_y F\right]$$

$$+ \frac{\alpha_1 h^3}{6}\left[3u_1^2 F_{xx} + 3\lambda_1^2 F^2 F_{yy} + 6\lambda_1 F u_1 F_{xy}\right]$$

$$+ \alpha_2 hF + \frac{\alpha_2 h^2}{2}\left[2F_x u_2 + 2F_y \lambda_2 F\right]$$

$$+ \frac{\alpha_2 h^3}{6}\left[3u_2^2 F_{xx} + 3F^2 \lambda_2^2 F_{yy} + 6\lambda_2 F u_2 F_{xy}\right]$$

$$y(x+h) = y(x) + hF + \frac{h^2}{2}\left(F_x + FF_y\right)$$

$$+ \frac{h^3}{6}\left(F_{xx} + 2F_{xy}F + F_y F_x + FF_y^2 + F^2 F_{yy}\right)$$

$$(\alpha_0 + \alpha_1 + \alpha_2) hf = hf$$

$$(\alpha_0 + \alpha_1 + \alpha_2) = 1 \quad (1)$$

$$\frac{h^2}{2}F_x = \frac{h^2}{2}F_x\left(2\alpha_1 u_1 + 2\alpha_2 u_2\right)$$

$$2\alpha_1 u_1 + 2\alpha_2 u_2 = 1 \quad (2)$$

$$\frac{h^2}{2}FF_y = \frac{h^2}{2}FF_y\left(2\lambda_1^{\alpha_1} + 2\lambda_2^{\alpha_2}\right)$$

$$\left(2\lambda_1^{\alpha_1} + 2\lambda_2^{\alpha_2}\right) = 1$$

$$\left(2\lambda_1\alpha_1 + 2\lambda_2\alpha_2\right) = 1 \quad (3)$$

$$\frac{h^3}{6}F_{xx} = \left(3u_1^2\alpha_1 + 3u_2^2\alpha_2\right)\frac{h^3}{6}F_{xx}$$

$$\left(3u_1^2\alpha_1 + 3u_2^2\alpha_2\right) = 1 \quad (4)$$

$$\frac{h^3}{6}\frac{2}{2}FF_{xy} = \frac{1}{2}FF_{xy}\frac{h^3}{6}\left(\alpha_1 6\lambda_1 u_1 + \alpha_2 6\lambda_2 u_2\right)$$

$$3\alpha_1\lambda_1 u_1 + 3\alpha_2\lambda_2 u_2 = 1 \quad (5)$$

$$\frac{h^3}{6}FF_{xy} = \frac{h^3}{6}F^2F_{yy}\left(\alpha_1 3\lambda_1^2 + \alpha_2 3\lambda_2^2\right)$$

$$3\alpha_1\lambda_1^2 + 3\alpha_2\lambda_2^2 = 1 \quad (6)$$

equations:

$$(\alpha_0+\alpha_1+\alpha_2) = 1 \quad (1)$$

$$\alpha_1 u_1 + \alpha_2 u_2 = 1/2 \quad (2)$$

$$\lambda_1 \alpha_1 + \lambda_2 \alpha_2 = 1/2 \quad (3)$$

$$u_1^2 \alpha_1 + u_2^2 \alpha_2 = 1/3 \quad (4)$$

$$\alpha_1 \lambda_1 u_1 + \alpha_2 \lambda_2 u_2 = 1/3 \quad (5)$$

$$\alpha_1 \lambda_1^2 + \alpha_2 \lambda_2^2 = 1/3 \quad (6)$$

$$\alpha_1 u_1 = \frac{1}{2} - \alpha_2 u_2$$

$$\alpha_1 = \frac{1}{2u_1} - \frac{\alpha_2 u_2}{u_1} \quad (2)$$

$$\alpha_1 = \frac{1}{3u_1^2} - \frac{u_2^2 \alpha_2}{u_1^2} \quad (4)$$

(2) - (4)

$$0 = \frac{1}{2u_1} - \frac{1}{3u_1^2} - \frac{\alpha_2 u_2}{u_1} + \frac{u_2^2 \alpha_2}{u_1^2}$$

$$0 = \frac{1}{2}u_1 - \frac{1}{3} - \alpha_2 u_2 u_1 + u_2^2 \alpha_2$$

$$\frac{1}{3} = \frac{1}{2}u_1 - \alpha_2 u_2 u_1 + u_2^2 \alpha_2$$

$$\frac{3}{2} = u_1 - 2\alpha_2 u_2 u_1 + 2u_2^2 \alpha_2$$

$$\frac{3}{2} = u_1 - 2\alpha_2 (u_2 u_1 - u_2^2)$$

$$\frac{2}{3} + 2\alpha_2 (u_2 u_1 - u_2^2) = u_1$$

$$2\alpha_2 (u_2 u_1 - u_2^2) = \frac{3u_2 - 2}{3}$$

$$\alpha_2 = \frac{3u_2 - 2}{6(u_2 u_1 - u_2^2)} \qquad \frac{2}{4} - \frac{1}{4}$$

$u_1 \neq u_2$

* still keep as free parameter *

$u_1 = 2$

$u_2 = 1$

$$\alpha_2 = \frac{3-1}{6(2-1)} = \frac{1}{2}$$

$$\alpha_1 = \frac{1}{2u_1} - \frac{\alpha_2 u_2}{u_1} = \frac{1}{2(1)} - \frac{1}{2}\frac{1}{2} = \frac{1}{4}$$

$\alpha_1 = 1/4, \quad \alpha_2 = 1/2, \quad \mu_1 = 1, \quad \mu_2 = 2$

$\alpha_0 + \frac{1}{4} + \frac{1}{2} = 1$

$\alpha_0 = \frac{4}{4} - \frac{2}{4} - \frac{1}{4} = \frac{1}{4}$

$\alpha_0 = \frac{1}{4}$

(6) $\alpha_1 \lambda_1^2 + \alpha_2 \lambda_2^2 = 1/3$

(3) $\lambda_1 \alpha_1 + \lambda_2 \alpha_2 = 1/2$

$\lambda_1 = \frac{1}{2\alpha_1} - \frac{\lambda_2 \alpha_2}{\alpha_1}$

$\alpha_1 \left(\frac{1}{2\alpha_1} - \frac{\lambda_2 \alpha_2}{\alpha_1}\right)^2 + \alpha_2 \lambda_2^2 = \frac{1}{3}$

$\alpha_1 \left(\frac{1}{4\alpha_1^2} - \frac{2\lambda_2 \alpha_2}{2\alpha_1^2} + \frac{\lambda_2^2 \alpha_2^2}{\alpha_1^2}\right) + \alpha_2 \lambda_2^2 = \frac{1}{3}$

(5) $\alpha_1 \left(\frac{1}{2\alpha_1} - \frac{\lambda_2 \alpha_2}{\alpha_1}\right) \mu_1 + \alpha_2 \lambda_2 \mu_2 = \frac{1}{3}$

$\frac{\alpha_1 \mu_1}{2\alpha_1} - \frac{\alpha_1 \alpha_2 \mu_1 \lambda_2}{\alpha_1} + \alpha_2 \lambda_2 \mu_2 = \frac{1}{3}$

$$\lambda_1 \alpha_1 + \lambda_2 \alpha_2 = \frac{1}{2}$$

$$\lambda_1 \alpha_1 u_1 + \lambda_2 u_2 \alpha_2 = 1/3$$

$$\lambda_1 \frac{1}{4} + \lambda_2 \frac{1}{2} = \frac{1}{2}$$

$$\lambda_1 \frac{1}{4} + \lambda_2 = \frac{1}{3}$$

$$\frac{\lambda_2}{2} - \lambda_2 = \frac{3}{6} - \frac{2}{6} = \frac{1}{6}$$

$$\frac{-\lambda_2}{2} = \frac{1}{6}$$

$$\lambda_2 = \frac{-1}{3} \qquad \lambda_1 = \frac{8}{3}$$

$\alpha_1 = 1/4,\ \alpha_2 = 1/2,\ u_1 = 1,\ u_2 = 2,\ \alpha_0 = 1/4,\ \lambda_2 = \frac{-1}{3},\ \lambda_1 = \frac{8}{3}$